Review Review Questions on MAT213 / MTH213

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Review Question 1

An object with initial temperature $T(0) = 100^\circ C$ is placed in a room with ambient temperature $T_a = 25^\circ C$. The rate of cooling follows Newton's Law of Cooling:

$$\frac{dT}{dt} = -k(T - T_a),$$

where $k > 0$ is a constant. Solve for $T(t)$ given that $T(0) = 100^\circ C$.

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Solution - Step 1: Separating Variables

Rearrange the equation:

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Solution - Step 2: Integrating and Solving for T

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Solution - Step 3: Applying Initial Condition

Using $T(0) = 100$:

$$100 = 25 + C_1 e^0.$$

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Since $e^0 = 1$, we obtain:

$$C_1 = 75.$$

Solution - Step 3: Applying Initial Condition

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$$100 = 25 + C_1 e^0.$$

Since $e^0 = 1$, we obtain:

$$C_1 = 75.$$

Thus, the temperature function is:

$$T(t) = 25 + 75e^{-kt}.$$

Solution - Step 3: Applying Initial Condition

Using $T(0) = 100$:

$$100 = 25 + C_1 e^0.$$

Since $e^0 = 1$, we obtain:

$$C_1 = 75.$$

Thus, the temperature function is:

$$T(t) = 25 + 75e^{-kt}.$$

Review Question 2

Solve the differential equation:

$$\frac{dy}{dx} + y = e^{-x}$$

using the integrating factor method.

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Solution to Review Question 2 - Step 1

The given equation is of the form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where $P(x) = 1$ and $Q(x) = e^{-x}$.

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The integrating factor (IF) is given by:

$$\mu(x) = e^{\int P(x)dx} = e^{\int 1dx} = e^{x}.$$

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Rewriting,

$$\frac{d}{dx}(e^x y) = 1.$$

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Multiplying through by e^x, we obtain:

$$e^x \frac{dy}{dx} + e^x y = 1.$$

Rewriting,

$$\frac{d}{dx}(e^x y) = 1.$$

Solution to Review Question 2 - Step 3

Integrating both sides:

$$e^x y = \int 1 dx = x + C.$$

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Thus,

$$y = e^{-x}(x + C).$$

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Solution to Review Question 3 - Step 2

This gives:

$$\frac{y^2}{2} = \frac{x^2}{2} + C.$$

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Multiplying by 2:

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Solution to Review Question 3 - Step 3

Taking the square root:

$$y = \pm\sqrt{x^2 + 2C}.$$

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$$(2xy + 3x^2)dx + (x^2 + 4y)dy = 0.$$

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Solution to Review Question 4 - Step 1: Checking Exactness

The given equation is in the form:

$$M(x, y)dx + N(x, y)dy = 0,$$

where $M(x, y) = 2xy + 3x^2$ and $N(x, y) = x^2 + 4y$.

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Compute $\dfrac{\partial M}{\partial y}$:

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Compute $\dfrac{\partial M}{\partial y}$:

$$\frac{\partial}{\partial y}(2xy + 3x^2) = 2x.$$

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$$\frac{\partial}{\partial y}(2xy + 3x^2) = 2x.$$

Compute $\dfrac{\partial N}{\partial x}$:

$$\frac{\partial}{\partial x}(x^2 + 4y) = 2x.$$

Since $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$, the equation is exact.

Solution to Review Question 4 - Step 2: Finding the Potential Function

Since the equation is exact, there exists a function $\Psi(x, y)$ such that:

$$\frac{\partial \Psi}{\partial x} = M = 2xy + 3x^2.$$

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Integrate with respect to x:

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Since the equation is exact, there exists a function $\Psi(x, y)$ such that:

$$\frac{\partial \Psi}{\partial x} = M = 2xy + 3x^2.$$

Integrate with respect to x:

$$\Psi(x, y) = \int (2xy + 3x^2)dx = x^2y + x^3 + g(y).$$

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Integrate with respect to x:

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Now, differentiate with respect to y and set equal to N:

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$$\frac{\partial \Psi}{\partial y} = x^2 + g'(y) = x^2 + 4y.$$

Thus, $g'(y) = 4y$, so integrating gives:

$$g(y) = 2y^2 + C.$$

Solution to Review Question 4 - Step 3: General Solution

Thus, the potential function is:

$$\Psi(x, y) = x^2y + x^3 + 2y^2 + C.$$

Solution to Review Question 4 - Step 3: General Solution

Thus, the potential function is:

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The solution to the differential equation is:

$$\Psi(x, y) = C \quad \Rightarrow \quad x^2y + x^3 + 2y^2 = C.$$

Review Question 5

Verify that the function $y(x) = x\cos(\ln x)$ is a solution of the differential equation:

$$x^2y'' - xy' + 2y = 0.$$

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$$y' = \frac{d}{dx}\left(x\cos(\ln x)\right) = \cos(\ln x) + x\cdot\left(-\sin(\ln x)\cdot\frac{1}{x}\right).$$

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Simplify:

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Simplify:

$$y'' = -\frac{\sin(\ln x)}{x} - \frac{\cos(\ln x)}{x}.$$

Solution - Step 3: Compute $x^2y'' - xy' + 2y$

Compute x^2y'':

$$x^2y'' = x^2\left(-\frac{\sin(\ln x)}{x} - \frac{\cos(\ln x)}{x}\right) = -x\sin(\ln x) - x\cos(\ln x).$$

Solution - Step 3: Compute $x^2y'' - xy' + 2y$

Compute x^2y'':

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Compute $-xy'$:

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Solution - Step 3: Compute $x^2y'' - xy' + 2y$

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Compute $2y$:

$$2y = 2x\cos(\ln x).$$

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Summing all terms:

$$x^2y''-xy'+2y = (-x\sin(\ln x)-x\cos(\ln x))+(-x\cos(\ln x)+x\sin(\ln x))+2x\,co$$

Solution - Step 3: Compute $x^2y'' - xy' + 2y$

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Solution - Step 4: Conclusion

Simplifying:

$$(-x\sin(\ln x)+x\sin(\ln x))+(-x\cos(\ln x)-x\cos(\ln x)+2x\cos(\ln x))=0.$$

Solution - Step 4: Conclusion

Simplifying:

$$(-x\sin(\ln x) + x\sin(\ln x)) + (-x\cos(\ln x) - x\cos(\ln x) + 2x\cos(\ln x)) = 0.$$

Thus, $x^2y'' - xy' + 2y = 0$, confirming that $y(x) = x\cos(\ln x)$ is indeed a solution.

Review Question 6

Find the value of the constant k for which the following differential equation is exact, and then solve it:

$$(kx + y)dx + (x + 3y)dy = 0.$$

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Solution - Step 1: Identifying M and N

The given equation is of the form:

$$M(x, y)dx + N(x, y)dy = 0,$$

where:

$$M(x, y) = kx + y, \quad N(x, y) = x + 3y.$$

Solution - Step 1: Identifying M and N

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To check exactness, compute $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$.

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To check exactness, compute $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$.

Solution - Step 2: Checking Exactness

Compute $\dfrac{\partial M}{\partial y}$:

$$\frac{\partial}{\partial y}(kx + y) = 1.$$

Solution - Step 2: Checking Exactness

Compute $\frac{\partial M}{\partial y}$:

$$\frac{\partial}{\partial y}(kx + y) = 1.$$

Compute $\frac{\partial N}{\partial x}$:

$$\frac{\partial}{\partial x}(x + 3y) = 1.$$

Solution - Step 2: Checking Exactness

Compute $\frac{\partial M}{\partial y}$:

$$\frac{\partial}{\partial y}(kx + y) = 1.$$

Compute $\frac{\partial N}{\partial x}$:

$$\frac{\partial}{\partial x}(x + 3y) = 1.$$

For the equation to be exact, we require:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Solution - Step 2: Checking Exactness

Compute $\frac{\partial M}{\partial y}$:

$$\frac{\partial}{\partial y}(kx + y) = 1.$$

Compute $\frac{\partial N}{\partial x}$:

$$\frac{\partial}{\partial x}(x + 3y) = 1.$$

For the equation to be exact, we require:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Since both derivatives are equal for all k, the equation is always exact for any value of k.

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Compute $\frac{\partial N}{\partial x}$:

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Solution - Step 3: Finding the Potential Function $\Psi(x, y)$

Since the equation is exact, there exists a function $\Psi(x, y)$ such that:

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Since the equation is exact, there exists a function $\Psi(x, y)$ such that:

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Integrate with respect to x:

$$\Psi(x, y) = \int (kx + y)dx = \frac{k}{2}x^2 + xy + g(y).$$

Solution - Step 3: Finding the Potential Function $\Psi(x, y)$

Since the equation is exact, there exists a function $\Psi(x, y)$ such that:

$$\frac{\partial \Psi}{\partial x} = M = kx + y.$$

Integrate with respect to x:

$$\Psi(x, y) = \int (kx + y)dx = \frac{k}{2}x^2 + xy + g(y).$$

Differentiate with respect to y and set equal to N:

$$\frac{\partial \Psi}{\partial y} = x + g'(y) = x + 3y.$$

Solution - Step 3: Finding the Potential Function $\Psi(x, y)$

Since the equation is exact, there exists a function $\Psi(x, y)$ such that:

$$\frac{\partial \Psi}{\partial x} = M = kx + y.$$

Integrate with respect to x:

$$\Psi(x, y) = \int (kx + y)dx = \frac{k}{2}x^2 + xy + g(y).$$

Differentiate with respect to y and set equal to N:

$$\frac{\partial \Psi}{\partial y} = x + g'(y) = x + 3y.$$

Solving for $g(y)$:

$$g'(y) = 3y \Rightarrow g(y) = \frac{3}{2}y^2 + C.$$

Solution - Step 3: Finding the Potential Function $\Psi(x, y)$

Since the equation is exact, there exists a function $\Psi(x, y)$ such that:

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Integrate with respect to x:

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Solving for $g(y)$:

$$g'(y) = 3y \Rightarrow g(y) = \frac{3}{2}y^2 + C.$$

Solution - Step 4: General Solution

Thus, the potential function is:

$$\Psi(x, y) = \frac{k}{2}x^2 + xy + \frac{3}{2}y^2 + C.$$

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$$\Psi(x,y) = \frac{k}{2}x^2 + xy + \frac{3}{2}y^2 + C.$$

The solution to the differential equation is:

$$\Psi(x,y) = C \quad \Rightarrow \quad \frac{k}{2}x^2 + xy + \frac{3}{2}y^2 = C.$$

Solution - Step 4: General Solution

Thus, the potential function is:

$$\Psi(x,y) = \frac{k}{2}x^2 + xy + \frac{3}{2}y^2 + C.$$

The solution to the differential equation is:

$$\Psi(x,y) = C \quad \Rightarrow \quad \frac{k}{2}x^2 + xy + \frac{3}{2}y^2 = C.$$

Review Question 7

Determine whether the set of functions $\{f_1(x), f_2(x)\}$ is linearly dependent or independent using the Wronskian approach.

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Solution - Step 1: Compute the Wronskian

Given functions $f_1(x)$ and $f_2(x)$, the Wronskian is defined as:

$$W(f_1, f_2)(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ f_1'(x) & f_2'(x) \end{vmatrix}.$$

Solution - Step 1: Compute the Wronskian

Given functions $f_1(x)$ and $f_2(x)$, the Wronskian is defined as:

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Compute the determinant:

$$W(f_1, f_2)(x) = f_1(x)f_2'(x) - f_2(x)f_1'(x).$$

Solution - Step 1: Compute the Wronskian

Given functions $f_1(x)$ and $f_2(x)$, the Wronskian is defined as:

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Compute the determinant:

$$W(f_1, f_2)(x) = f_1(x)f_2'(x) - f_2(x)f_1'(x).$$

Solution - Step 2: Analyze the Wronskian

If $W(f_1, f_2)(x) \neq 0$ for some x, the functions are linearly independent.

Solution - Step 2: Analyze the Wronskian

If $W(f_1, f_2)(x) \neq 0$ for some x, the functions are linearly independent. If $W(f_1, f_2)(x) = 0$ for all x, the functions are linearly dependent.

Solution - Step 2: Analyze the Wronskian

If $W(f_1, f_2)(x) \neq 0$ for some x, the functions are linearly independent. If $W(f_1, f_2)(x) = 0$ for all x, the functions are linearly dependent.

Solution - Step 3: Example

Consider $f_1(x) = e^x$ and $f_2(x) = e^{-x}$.

Solution - Step 3: Example

Consider $f_1(x) = e^x$ and $f_2(x) = e^{-x}$. Compute derivatives:

$$f_1'(x) = e^x, \quad f_2'(x) = -e^{-x}.$$

Solution - Step 3: Example

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